Attacks in Gossip Learning

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1 INTRODUCTION

The rapid expansion of Artificial Intelligence (AI) has transformed the way data is collected and processed, driving the adoption of cloud-based AI solutions for more processes. In traditional machine learning, storing training data has always been the heaviest part of the process. Companies must go through a long and costly process of gathering all user data in a single machine or in a data center, increasing potential bottlenecks and privacy concerns.

In an effort to overcome these limitations, Federated Learning (FL) [1] was presented by Google. The idea was that instead of gathering all the raw data, each decentralized device could train their own model and then aggregate all the parameters on a central server, thus mitigating some of the inherent privacy risks and reducing the cost of data storage. However, despite being a major step forward, FL still presents significant privacy risks, particularly regarding the central server's role in coordination and control.

More recently, Decentralized Federated Learning (DFL) [2] [3], also known as Gossip Learning (GL), has been proposed as an even more decentralized approach. By enabling peer-to-peer model updates and information exchange, gossip learning further minimizes the dependence on a centralized infrastructure and distributes the training process across a network of devices. Nevertheless, the decentralized nature of GL also brings unique security challenges. In this paper, we explore these security risks in depth, examining how various attacks can compromise gossip learning systems.

The attacks discussed will be categorized into two main groups: privacy attacks and utility attacks. This classification highlights the specific vulnerabilities of GL and provides a clearer understanding of the security challenges it faces. By analyzing these threats, we aim to shed light on the risks associated with decentralized learning and emphasize the need for robust defense mechanisms.

2 BACKGROUND

Many attacks on Gossip Learning are adaptations of existing attack strategies in other learning paradigms. However, to better comprehend the risks and vulnerabilities that those attacks carry, it is crucial to have a clear understanding of the fundamentals of Federated Learning and Gossip Learning. Thus, the following section provides a concise overview of these concepts, highlighting their key mechanisms and differences.

2.1 Federated Learning

Before the advent of FL, collaborative training required the extensive exchange of data between users, leading to a centralized repository of training data. This approach, however, had significant challenges, including users' continuous communication, high risks of data leakage, and potential communication bottlenecks.

Federated Learning was developed to address these issues. In FL, a central server distributes a common model to all participating users. Each user trains the model locally using their own data, and once the local model converges, its parameters are sent back to the central server. The server then aggregates these parameters using a specified algorithm and distributes the updated model back to the users. This process is iteratively repeated until the global model converges.

FL initially assumes homogeneous data across users, but real-world scenarios often involve heterogeneous data. To address this, FL has evolved into three main categories: Horizontal Federated Learning (HFL), Vertical Federated Learning (VFL), and Federated Transfer Learning (FTL). HFL is used when data across users is similar in features but differs in user-specific samples, typically seen in mobile devices. VFL applies when users have different features but the same set of samples, such as demographic and purchase data. FTL combines FL with transfer learning to handle heterogeneous data in both features and distributions, improving learning efficiency when data is sparse or unbalanced.

Despite the advantages of FL, several challenges remain [4] . One of the main concerns is the reliance on a central server for aggregating model parameters, which can create a single point of failure and bottleneck, especially when scaling to a large number of participants. Additionally, the communication overhead involved in transmitting model updates can be significant, particularly when data is heterogeneous across users. While FL some methods address some of these challenges, they still struggle with issues like data imbalance, privacy concerns, and limited scalability. These persistent challenges underscore the need for more decentralized and robust frameworks, such as Gossip Learning, which aim to eliminate the central server and reduce communication costs, while further enhancing privacy and scalability.

2.2 Gossip Learning

Knowing that most of the challenges and risks of FL come from the central server, a Decentralized Federated Learning was proposed, a system that works by peer-to-peer communication and has no server connected to all users.

With this new method, also known as Gossip Learning, the users with data are defined as nodes that communicate with each other to improve the model performance. Each node has its own model that is trained with the local data, once it converges, the model parameters are sent to different nodes. When parameters arrive to a new node, local parameters are aggregated with the new ones and the model is trained again with the updated parameters. This process is iteratively repeated until the global model converges.

Despite its promising approach to decentralization, GL also comes with challenges. With the absence of a central server some privacy issues and risks can be removed, but some new risks should be taken into account since that server was also the central authority that validated the updates and established trust and verification among nodes.

The following sections will dive deeper into the specific security vulnerabilities of Gossip Learning, categorizing the attacks into privacy and utility types and exploring their underlying mechanisms.

3 PRIVACY ATTACKS

One of the primary concerns is the risk of privacy attacks. Since each node independently shares model parameters with its peers, there is a possibility for adversaries to intercept these communications and reconstruct sensitive information about the local data. An overview of these attacks for GL is given in the following.

3.1 Inference Attacks

Inference attacks aim to extract private information from exchanged model parameters without directly accessing the raw data. These attacks exploit patterns in model updates to reveal whether specific data samples were used for training or to infer sensitive properties of the dataset. GL is particularly vulnerable to such attacks, as nodes frequently share model parameters in a decentralized manner, creating multiple opportunities for adversaries to intercept updates and extract private information.

Since the parameters or gradients in GL are derived from clients' private data, analyzing model updates can reveal substantial information about individual participants.

1) *Membership inference attacks:* membership inference attacks aim to determine whether a specific data sample was part of the training dataset [5]. This attack takes advantage of the fact that models often behave differently when encountering data they have seen during training compared to unseen data.

For instance, in a neural network trained on natural language tasks, non-zero gradients in the embedding layer may expose which tokens appeared in the benign clients' training batches. By observing model updates over multiple rounds, an attacker can deduce whether a particular word, image, or record was used for training. In the context of Gossip Learning, this means an adversary could determine whether a particular individual's sensitive information—such as a medical record—was part of the training data.

2) *Class representation inference attacks:* rather than targeting individual samples, class representation inference attacks attempt to reconstruct the general characteristics of data belonging to a specific class in the training dataset. By analyzing how the model updates change over time, an attacker can infer the statistical properties of each class, such as common patterns or feature distributions.

For example, if GL is being used to train a model for handwritten digit recognition, an adversary could infer structural patterns that define each digit class. Similarly, in a financial fraud detection system, an attacker might deduce transaction patterns that are indicative of fraudulent behavior, even without accessing raw transaction data.

3) *Property Inference Attacks:* in property inference attacks, adversaries seek to learn sensitive attributes about the training dataset, even if those attributes are not explicitly used for classification. Unlike membership inference, which aims to determine whether a sample was included in training, property inference attacks attempt to extract statistical properties of the data.

For instance, consider a GL system trained on patient records for disease prediction. An attacker might not be able to identify individual patient records but could still infer that a certain percentage of the training data corresponds to patients with a specific genetic predisposition. This can be highly problematic in settings where privacy is paramount, such as healthcare or finance.

3.2 Model Inversion Attacks

In a GL environment, where nodes exchange model parameters rather than raw data, attackers can leverage Model Inversion Attacks (MIAs) to extract private information. These attacks can go beyond simple data leakage to reconstruct precise details of the training data used by other nodes [6] .

White-box scenarios are the most common in GL, the attacker has full access to the model architecture and parameters. This privileged access enables the use of several inference techniques, such as Membership Inference Attacks or gradient-based reconstruction methods. Gradient ascent is one of the most effective strategies used in MIAs. In this technique, the attacker leverages the received gradients to iteratively adjust input data. The goal is to maximize the model's output in a way that reconstructs sensitive data from the model's parameters, essentially reversing the model's decision-making process to infer what data may have been used in training.

The success of MIAs, particularly gradient-based methods, depends on factors such as the model's complexity, the richness of the training data, and the attacker's access to model parameters. While gradient ascent can be effective, its success may vary based on these factors.

3.3 GAN Reconstruction Attacks

GAN-based attacks leverage Generative Adversarial Networks (GANs) to reconstruct private training data from intercepted model updates [7] [8]. Unlike inference attacks, which only aim to infer properties of the training data, GAN reconstruction attacks generate realistic samples that closely resemble the original data used by benign clients. This poses a significant privacy threat in Gossip Learning, where model parameters are frequently exchanged between nodes in a decentralized manner.

In this type of attack, an adversary intercepts multiple rounds of model updates and utilizes the extracted features to train a GAN. The attacker exploits patterns in the intercepted gradients or parameters to approximate the original training data. The GAN consists of two competing neural networks: a generator and a discriminator. The generator attempts to produce synthetic samples that mimic the real training data, while the discriminator tries to distinguish between the synthetic samples and real representations reconstructed from the extracted model features. These real representations can be derived using optimization techniques such as gradient inversion or gradient ascent, which attempt to reverse-engineer data samples from the intercepted model updates. (jada real representationak badauzkagu gradient ascend eginda zertako entrenatu GAN? adibide gehiago lortzeko bakarrik?)

As the training process progresses, the generator refines its outputs by learning to produce increasingly realistic data representations, while the discriminator improves its ability to differentiate between genuine and synthetic samples. The adversary continues this adversarial training process until the discriminator is unable to reliably distinguish between real and fake samples, at which point the generator has successfully learned to reconstruct plausible representations of the private training data. This method enables

an attacker to obtain visually and statistically similar samples to those used in training, significantly compromising the privacy of decentralized learning systems.

3.4 Man-in-the-middle Attacks

In a GL environment, a Man-in-the-Middle (MitM) attack involves an attacker secretly intercepting, modifying, or injecting malicious data into the communication between two or more nodes during the model exchange process. Unlike traditional attacks that focus solely on the direct data or model parameters, MitM attacks occur during the transmission phase and aim to manipulate the information being shared among nodes without the participants' knowledge [7].

These attacks pose a threat to the privacy of the participating nodes. An attacker could manipulate the data exchanged during the gossip process in a way that inadvertently exposes private information about the nodes' training data, similar to model inversion attacks. For example, by controlling the model updates, the attacker could potentially extract or infer sensitive information about the local data used by other nodes in the network.

Although this attack has been classified as privacy attack, model utility can also be easily affected. The attacker may alter the model parameters or gradients being transmitted between the nodes, either to inject errors into the global model or to inject harmful, poisoned data that negatively impacts the overall learning. For instance, a malicious node could listen in on the model updates and either modify the parameters (e.g., changing the weights of a neural network) or inject false updates that mislead the global aggregation process. These altered updates could make the global model converge to a suboptimal or biased state, corrupting the training process and reducing the model's utility.

One of the challenges posed by MitM attacks in GL is that they are often difficult to detect. Since GL is decentralized, there is no central server or authority to oversee or validate the model exchanges. As a result, it is easy for an attacker to intercept the communication between honest nodes without raising suspicion. Furthermore, because the attacker does not need to have access to the raw data itself (only the model updates being shared), the attack becomes more feasible in decentralized settings where nodes only exchange model parameters and not sensitive data.

4 UTILITY ATTACKS

In addition to privacy risks, GL is also vulnerable to utility attacks. Malicious nodes might deliberately inject misleading parameters or corrupt the model updates, thereby degrading the overall performance of the global model. Such attacks could result in a model that converges to a suboptimal or even harmful state. The challenge is further compounded by the difficulty of detecting and isolating these malicious actors without a central oversight mechanism.

4.1 Poisoning Attacks

Poisoning attacks are one of the most significant threats to the utility and reliability of a machine learning system, particularly in decentralized systems like GL. In a poisoning attack, an adversary intentionally manipulates the training process by injecting malicious updates into the model. These attacks can occur at various stages of the model training process, such as during the transmission of model updates or the aggregation phase, and their goal is to degrade the quality or accuracy of the global model.

In the context of GL, the impact of poisoning attacks is dependent on several factors, including the number of nodes the attacker controls and the number of connections those nodes have to other nodes in the network. If the attacker controls a significant portion of the network or can influence highly connected nodes, the attacker may be able to inject poisoned updates that have a more significant impact on the global model. On the other hand, if the attacker controls only a small number of nodes or nodes with fewer connections, the injected malicious updates may be less influential, as the global model aggregates updates from a larger pool of participants. (Hau ez dakit beti egia den edo ez)

However, the decentralized nature of GL makes it more difficult to detect and isolate such attacks compared to centralized systems. The gradual aggregation of model updates means that even small, malicious modifications to the model parameters can accumulate over time, subtly degrading the global model's performance or even causing it to converge to a suboptimal or biased state.

The poisoning attacks could take different forms:

4.1.1 Data Poisoning Attacks. In a poisoning attack, the training data is modified instead of directly attacking model parameters. False or strategically changed data is send to the model to affect the utility. We can divide the data poisoning attacks into two major categories: clean label attacks and dirty label attacks [9] .

1) *Clean label attacks:* in clean label attacks, the attacker manipulates the input features of training samples, while the labels remain unchanged. This type of attack assumes the attacker cannot alter the labels due to a verification process. However, in the context of GL, where data verification processes are often weaker than in centralized systems like Federated Learning (FL), clean label attacks are less common but still a potential risk. (GL batean verification process potente bat egon daiteke?)

2) *Dirty label attacks:* in contrast, dirty label attacks target the labels of the training data. In situations where verification is either absent or easily bypassed, these attacks—such as label flipping—become more effective. By altering labels, the attacker can mislead the model, causing it to learn incorrect associations and adversely impacting its performance.

4.1.2 Model Poisoning Attacks. A more direct way to manipulate the client's local model parameters is a Model Poisoning Attack [10]. For instance, if an adversary controls a node in the network, they could intentionally change the weights of the neural network, introduce incorrect biases, or modify gradient updates in ways that are harmful to the model's performance. The objective is to influence the global model's aggregation process in ways that either degrade its performance or lead it to learn undesirable behaviors. In extreme cases, this type of attack could even lead the model to learn malicious behavior, rendering the model useless or dangerous.

4.1.3 Backdoor Poisoning Attack. Adversaries can manipulate the training data by modifying individual features or adding watermarks to small regions of the original training dataset [7]. Backdoors embedded in the model will act on triggers in the inputs (e.g.,

watermarks, a stamp on images). Meanwhile, the accuracy of the poisoned model on clean data remains untouched.

4.2 Evasion Attacks

Evasion attacks target the ability of the global model to generalize well, particularly during its deployment phase. In a typical evasion attack, an adversary modifies its input data to trick the model into making incorrect predictions or classifications [7]. In the case of GL, where the model is aggregated from various decentralized nodes, an attacker might alter their local data to exploit weaknesses in the global model. Although evasion attacks are usually done in test time, by submitting malicious inputs during the training phase, attackers can influence the global model's decision-making process. (Poisoning attack ¿?¿?).

4.3 DDoS Attacks

Distributed Denial of Service (DDoS) attacks [11] in the context of Gossip Learning primarily aim to disrupt the normal functioning of the GL system by overwhelming it with malicious or excessive requests. In GL, nodes exchange model parameters over the network, and these exchanges are crucial for the learning process. A DDoS attack could target the communication channel, flooding the network with an excessive number of requests or fake updates, thereby disrupting the flow of legitimate updates and hindering the convergence of the global model.

The goal of a DDoS attack in a GL system is to exhaust the network's bandwidth or computational resources, making it difficult for honest nodes to communicate effectively. This could result in prolonged training times, delayed convergence, or even complete system failure. Since GL is decentralized, detecting and mitigating DDoS attacks can be particularly challenging. Without a central server to manage and validate the nodes' activity, identifying malicious nodes or abnormal traffic patterns may require sophisticated techniques to isolate and block the attacking nodes.

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